Haotian Ma

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INTRODUCTION

Computer Science Ph.D. candidate at the University of Wisconsin–Madison, developing methods to understand how learned representations shape model behavior across interventions, combinations, and changing contexts. My work combines theory, method development, controlled interventions, open-source research infrastructure, and large-scale evaluation across **mechanistic interpretability**, **LLM reasoning**, **AI safety**, and **AI for scientific discovery**.

EDUCATION

University of Wisconsin–Madison — Madison, WI
Ph.D. in Computer Sciences — Sep 2021 – Jun 2027
- Advisor: Prof. Daifeng Wang

Southern University of Science and Technology — Shenzhen, China
B.S. in Physics — Sep 2016 – Jun 2021
- Advisor: Prof. Hu Xu

RESEARCH

Faithful Representations and Mechanistic Interpretability

Understand feature representations and how those representations support faithful explanations and interventions.

Sparse Autoencoders Do Not Guarantee Local Features — **ICLR 2027 Submission**
Haotian Ma [code]
- Demonstrated that accurate sparse autoencoder (SAE) reconstruction does not guarantee localized, interpretable features.
- Proved the same SAE can exhibit different input dependencies despite identical representation samples, and validated this across 5 trained SAEs.

Representation Redistribution in Sparse Autoencoders — **AAAI 2027 Submission**
Ziye Mei, ***Haotian Ma*** [paper] [code]
- Found that prompt changes preserving the same request can redistribute sparse feature activity, with patterns persisting across unseen prompt variants.
- Validated across 27 SAEs spanning Gemma, Llama, Mistral, GPT, and Pythia, and assessed how they helped predict failures in frozen safety probes.

Quantification and Analysis of Layer-wise and Pixel-wise Information Discarding — **ICML 2022 Spotlight**
Haotian Ma*, *Hao Zhang****, *Fan Zhou, Yinqing Zhang, Quanshi Zhang* [paper] [code]
- Introduced information entropy measures that compare how neural networks discard input information across layers, models, and individual pixels.
- Derived a theoretical link between selective information discarding and the efficiency of extracting features relevant to the task.

Interpreting CNNs via Decision Trees — **CVPR 2019**
Quanshi Zhang, Yu Yang, ***Haotian Ma***, *Ying Nian Wu* [paper]
- Converted hidden CNN features into object part concepts and quantified how those parts drive individual predictions.

Interpretable AI for Scientific Discovery

Develop interpretable methods that connect model decisions to scientifically meaningful spatial, spectral, and molecular structure.

Spectral-Spatial Interpretation — **NeurIPS 2026**
Haotian Ma, *Ruqi Yang, Philip Townsend* [paper] [code]
- Pioneered a framework that jointly identifies which spatial locations drive predictions and which spectral channels matter at each location.
- Outperformed Integrated Gradients, DeepSHAP, KernelSHAP, and RISE across 3 datasets in joint faithfulness, spatial localization, and spectral attribution.

SegJointGene: Joint Cell Segmentation and Spatial Gene Prioritization by Information Entropy Guided CNNs — **Bioinformatics 2026**
Haotian Ma, *Daifeng Wang* [paper] [code]
- Developed an information entropy framework for cell segmentation and gene prioritization, improving molecular signal assignment accuracy by 5–20%.
- Identified neuronal development and synaptic signaling genes in brain tissue and lineage markers in human tonsil.

Spatial Phenotyping Linking Single Cell Genomics to Disease Pathology through Joint Deep Representation Learning — **ISMB 2026 Poster**
Chenfeng He, ***Haotian Ma***, *Pubudu Kumarage, Xuerou Li, Kalpana Hanthanan Arachchilage, Shuang Liu, Daifeng Wang* [paper]
- Learned joint representations of disease phenotype, gene expression, and spatial organization to identify cellular niches associated with disease.
- Integrated single cell RNA sequencing with spatial transcriptomics to transfer phenotype signals across cohorts and map disease pathology.

Compositionality, Intervention, and Systematic Reasoning in LLMs

Study when model components and structural transformations can be combined, reused, or generalized reliably in unseen settings.

Behavioral Compositionality of Sparse Autoencoder Representations — **ICLR 2027 Submission**
Haotian Ma, *Ziye Mei*, *Frederic Sala* [paper] [code]
- Characterized downstream interactions among SAE features, revealing combined effects differ from the sum of individual feature effects.
- Pairwise modeling improved combined steering prediction in Llama-3.1 8B by 33%.

Certifying Reuse for LLM Components from Local Interventions — **ICLR 2027 Submission**
Haotian Ma, *Ruqi Yang*
- Formulated a framework to certify when locally tested LLM components can be reused in combinations never evaluated directly.
- Proved a few targeted tests can certify component reuse in untested combinations without evaluating every combination.

SymSelect: Learning Which Transformations to Trust for Systematic Reasoning — **NAACL 2027 Submission**
Haotian Ma, *Ruqi Yang* [code]
- Established representation similarity as a signal for selecting useful transformations without validity labels.
- Recovered 77 to 92% of supervised alignment gains on MNS and I-RAVEN across ResNet, ViT, CLIP, and Context Choice Transformer.

** Equal contribution*

Reliable LLMs and AI Safety

Characterize how AI systems remain reliable under uncertain preferences, underspecified user identities, and risks of private information leakage.

The Limits of Reward Guidance under Finite Adaptation **COLING 2027 Submission**

Haotian Ma [code]

- Derived an exact scalar support horizon for how long preference data can certify improvement under finite reward adaptation.
- Proved certification can fail under policy gradient despite pointwise feasible rewards; bounded labels needed to extend the horizon.

Gender Attribution by Proxy: Restraint, Fallback, and Rationalization in LLMs **AAAI 2027 Submission**

*Chenkun Jiang**, Haotian Ma**, Kai Chen, Haiyuan Liu*

- Introduced gender attribution by proxy, showing how LLMs turn ordinary user context into unsupported gender assumptions.
- Found that 6 LLMs systematically converted ordinary contextual cues into unsupported gender attributions when user gender was unspecified.

Interpretable Complex-Valued Neural Networks for Privacy Protection **ICLR 2020**

*Liyao Xiang, Hao Zhang, **Haotian Ma**, Yifan Zhang, Jie Ren, Quanshi Zhang* [paper]

- Designed neural networks that encode intermediate features as complex values with randomized phases while preserving accurate predictions.
- Reduced attackers' ability to reconstruct inputs or infer hidden properties across 8 neural networks.

EXPERIENCE

OpenSAE Madison, WI
Founder & Maintainer. Built research infrastructure for large-scale LLM evaluation, SAE interventions, and benchmarking. Oct 2026 – Present

Waisman Center, University of Wisconsin–Madison Madison, WI
Graduate Research Assistant. Led methodological development and quantitative analysis for interpretable methods in spatial omics. Sep 2023 – Present

- Advisor: Prof. Daifeng Wang

Department of Computer Sciences, University of Wisconsin–Madison Madison, WI
Graduate Research Assistant. Developed symbolic labeling rules and evaluation pipelines for automated weak supervision. Sep 2021 – Jun 2022

- Advisors: Prof. Aws Albarghouthi and Prof. Frederic Sala

Center for Vision, Cognition, Learning, and Autonomy (VCLA), UCLA Los Angeles, CA
Research Intern. Explored energy-based modeling and efficient sampling methods for protein discovery. Jul 2020 – Oct 2020

- Advisor: Prof. Ying Nian Wu

Department of Physics, Southern University of Science and Technology Shenzhen, China
Research Assistant. Applied generative models and efficient sampling methods to identify stable crystal structures. Sep 2019 – Jun 2020

- Advisor: Prof. Hu Xu

John Hopcroft Center, Shanghai Jiao Tong University Shanghai, China
Research Intern. Investigated interpretable methods to explain the prediction logic of convolutional neural networks. Jul 2018 – Jul 2019

- Advisor: Prof. Quanshi Zhang

SKILLS

- **LLMs & Agent Systems:** LLM Evaluation, LLM Inference, Prompt Engineering, Synthetic Data Generation, Multi-Agent Orchestration, Large-Scale Agent Experimentation, Evaluation Harnesses, vLLM
- **Machine Learning & Interpretability:** Mechanistic Interpretability, Sparse Autoencoders, Representation Analysis, Representation Interventions, LLM Steering, Feature Attribution, Multimodal Learning, Transformers
- **GPU, HPC & Performance Engineering:** CUDA/GPU, GPU Parallelism, Parallel Experiment Execution, Performance Optimization, Slurm/HPC, Linux, Bash, Docker
- **Statistical & Scientific Computing:** Statistical Inference, Hierarchical Bayesian Modeling, Low-Rank Modeling, Image Segmentation, Spatial Omics, Spatial Transcriptomics, Spatial Proteomics
- **Programming & Tools:** Python, C++, R, SQL, PyTorch, TensorFlow, scikit-learn, Git

AWARDS

- **Top Reviewer**, NeurIPS, 2026
- **Silver Reviewer**, ICML, 2026
- **Student and Young Professional Award**, ACM BCB, 2025
- **UW–Madison Computer Sciences Scholarship**, 2024
- **UW–Madison Computer Sciences First-Year Scholarship**, 2021
- **Second Prize**, ASC Student Supercomputer Challenge, 2018
- **Silver Medal**, International Genetically Engineered Machine (iGEM) Competition, 2018
- **Outstanding Participant, Ranked 6th of 120**, Nankai University National High School Philosophy Summer Program, 2015
- **National Second Prize**, 6th National Mathematics, Physics and Chemistry Competition for Secondary School Students, 2014

PROFESSIONAL SERVICE

- **Journal Reviewer:** Pattern Recognition (Elsevier) and IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- **Conference Program Committee:** NeurIPS 2026, ECCV 2026, ICML 2026, CVPR 2026, etc.

TEACHING EXPERIENCE

- **Teaching Assistant**, UW–Madison: Machine Learning (CS 760)
- **Teaching Assistant**, UW–Madison: Introduction to Artificial Intelligence (CS 540)
- **Teaching Assistant**, UW–Madison: Data Science Programming II (CS 320)
- **Education Assistant**, UW–Madison ALP Precollege: Introduction to Neuroscience